

Non-HIV Antiviral Agents

John F. Fisher, MD • 66-slide current deck • structure → activation → viral target → use → toxicity

3 official objectives

1 ScholarRx brick

Herpes antivirals

Influenza / RSV / COVID-19

HBV / HCV

SOURCE PRIORITY: the current 66-slide Fisher deck is primary. The attached handout is used only as secondary enrichment. Its final section on HIV antiretroviral therapy is explicitly labeled “Beyond the Slides” and is excluded from this resource because you asked to call this lecture Non-HIV Antiviral Agents and HIV pharmacology is tracked as a separate Week 8 core lecture.

Official Week 8 Objectives

1. Describe the structures of DNA and RNA.
2. Outline the mechanisms of action of common antiviral and antiretroviral drugs.
3. List the major side effects of antiviral drugs.

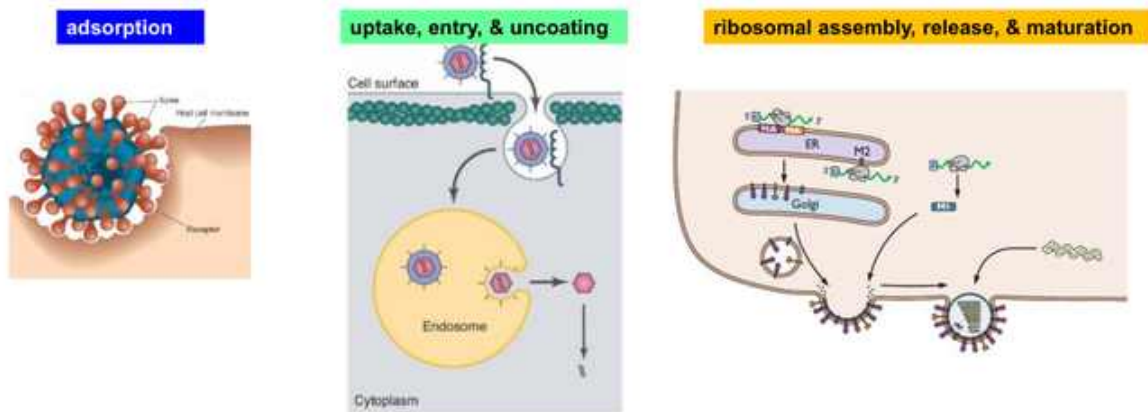
ASSIGNED SCHOLARRX

Antiviral Drugs: Foundations & Frameworks

Scope note: the official objective says antiviral *and antiretroviral* mechanisms, but the current Fisher deck is overwhelmingly non-HIV pharmacology. HIV ART remains a separate Week 8 lecture, so this companion does not import the handout's “Beyond the Slides” HIV section into the faculty content.

Slides 2-9: Antiviral Logic Starts With the Viral Life Cycle

Major targets of anti-virals

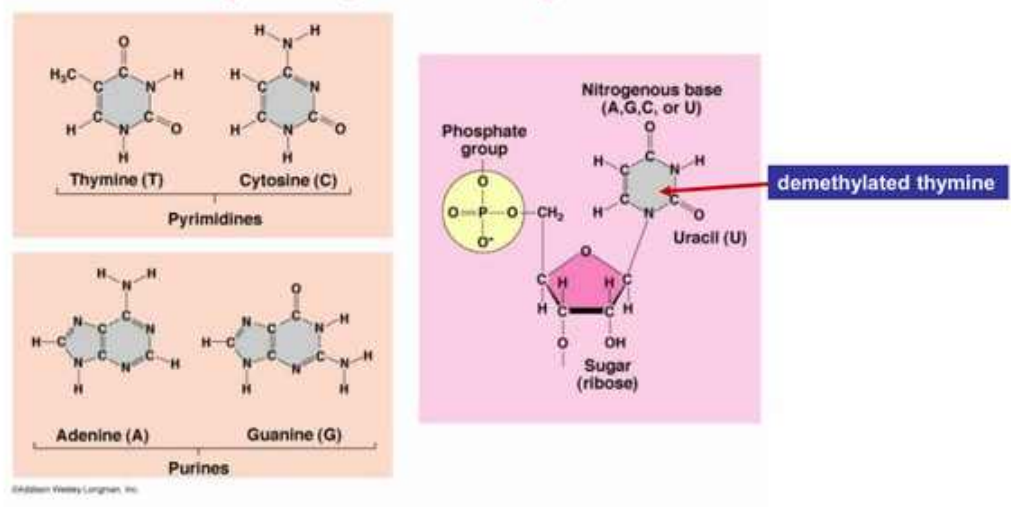


Slide 2. Major antiviral targets in the deck: adsorption, uptake/entry/uncoating, and assembly/release/maturation.

MECHANISM FRAMEWORK

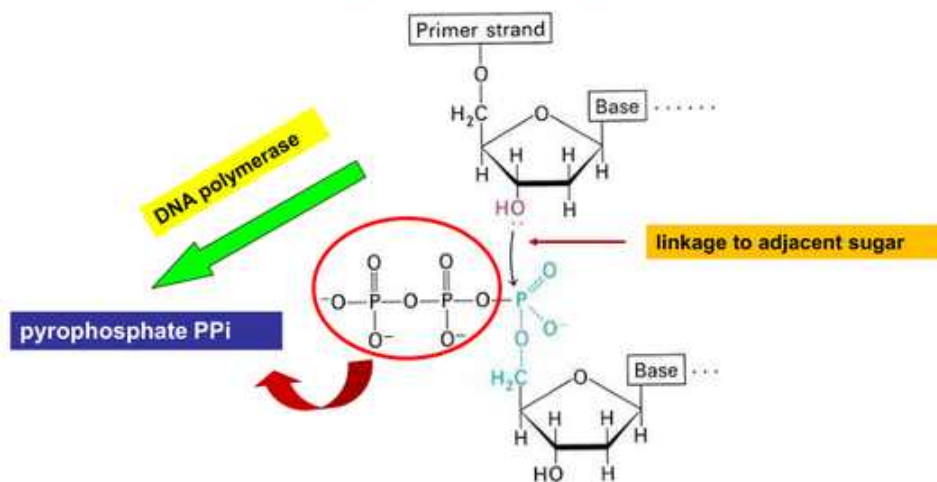
Block entry • corrupt genome synthesis • prevent protein processing/assembly • prevent release. Antivirals work best when they exploit a viral enzyme or replication step that differs from host-cell biology.

DNA & RNA (components)



Slide 4. DNA/RNA components. The lecture then walks through nucleic-acid structure before introducing nucleotide analog pharmacology.

DNA (formation)



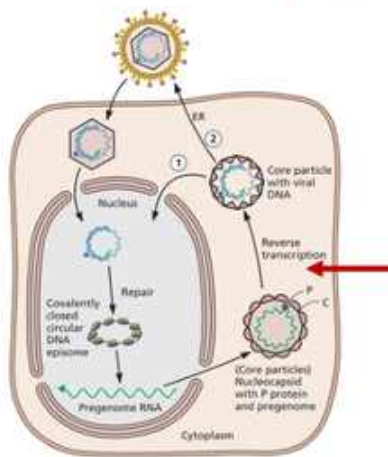
Slide 9. Normal DNA formation releases pyrophosphate as DNA polymerase links the next nucleotide. This chemistry becomes the basis for both chain-terminating analogs and foscarnet.

Handout-only clarification: nucleoside = base + sugar; nucleotide = nucleoside + phosphate(s). Many antivirals behave as “fake” nucleos(t)ides that are incorporated into viral nucleic acid and terminate elongation.

Slides 10-34: Herpesvirus Antivirals

Acyclovir / Valacyclovir / Famciclovir

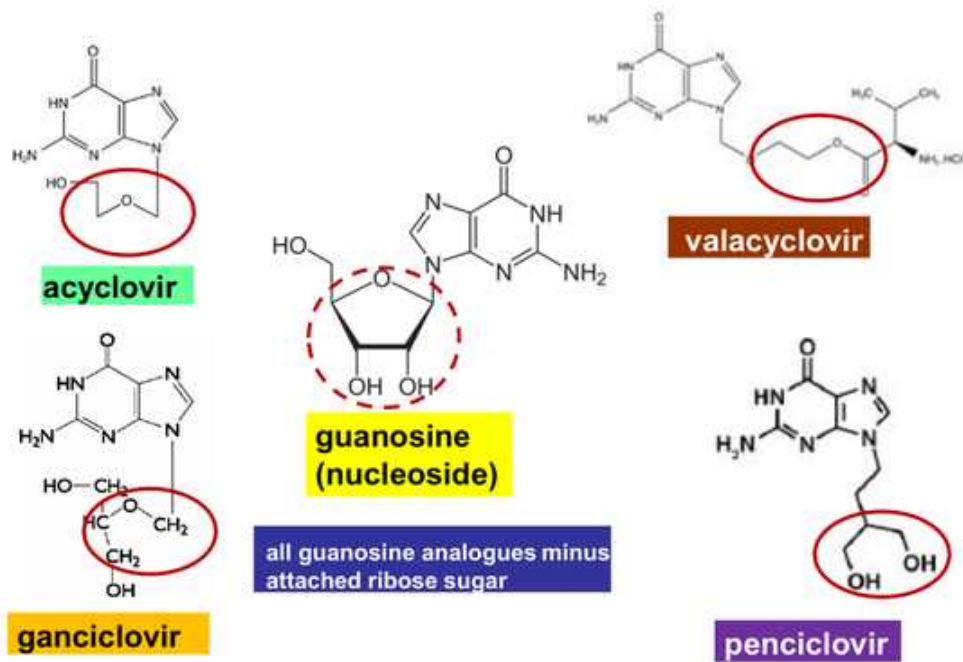
Herpes simplex virus (replication)



HSV uses thymidine kinase to replicate its DNA:

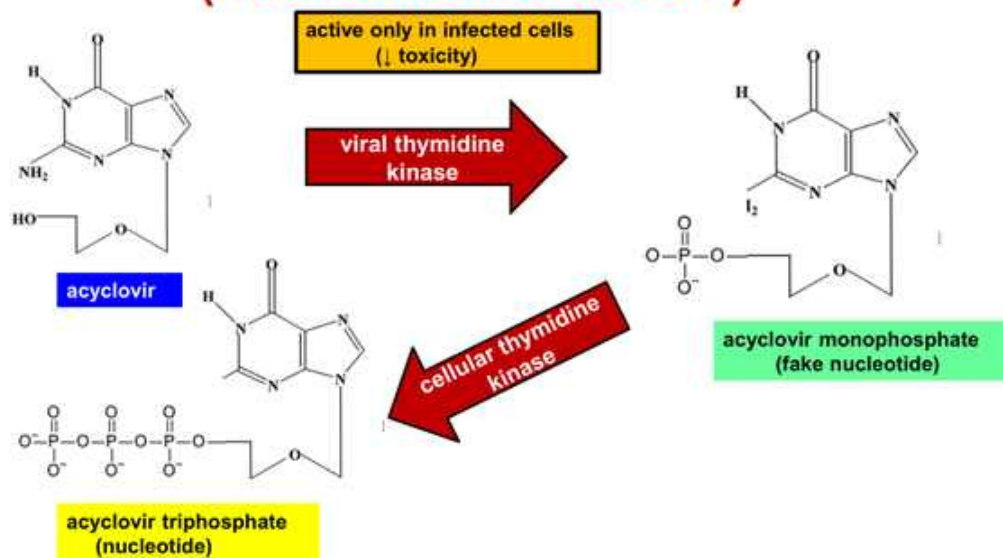
thymidine → thymidine mono PO₄ → thymidine triphosphate
OR
guanine → guanine mono PO₄ → guanine triphosphate

Slide 10. HSV uses viral thymidine kinase during DNA replication; this viral kinase becomes the selective activation step for acyclovir.



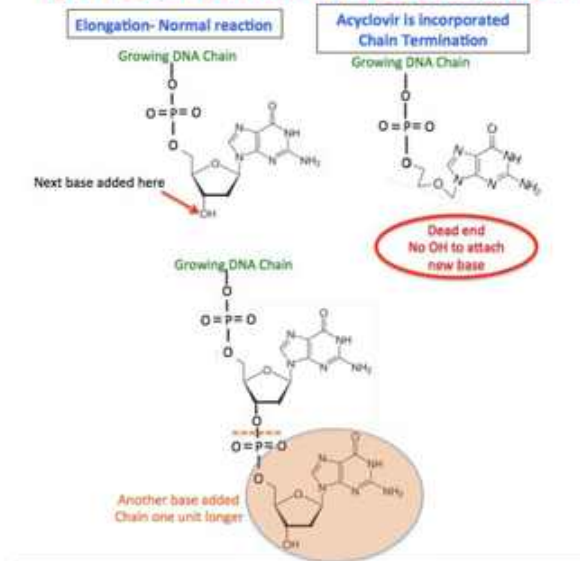
Slide 11. Acyclovir, ganciclovir, valacyclovir, and penciclovir are shown structurally as guanosine analogs lacking the normal attached ribose sugar.

Acyclovir (mechanism of action)



Slide 14. Acyclovir is first phosphorylated by viral thymidine kinase, then by cellular kinases. The slide emphasizes activity in infected cells and therefore lower toxicity.

Acyclovir (mechanism of action)

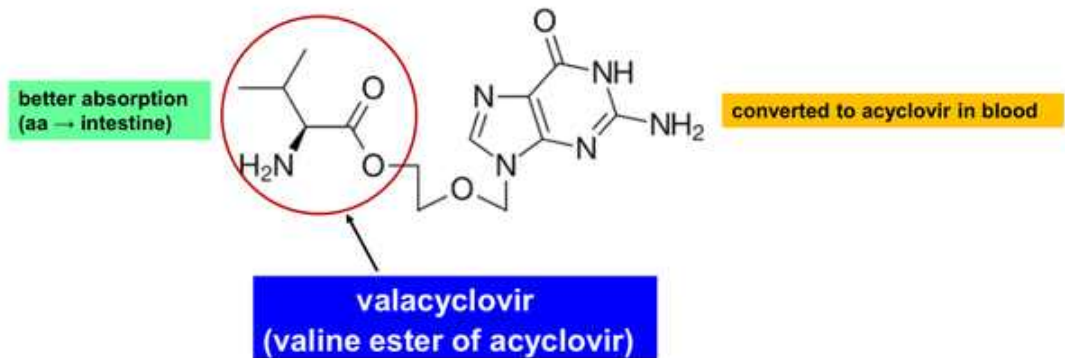


Slide 15. Once incorporated, acyclovir creates a dead end because the growing DNA chain lacks the OH needed to add the next base.

ACYCLOVIR MECHANISM

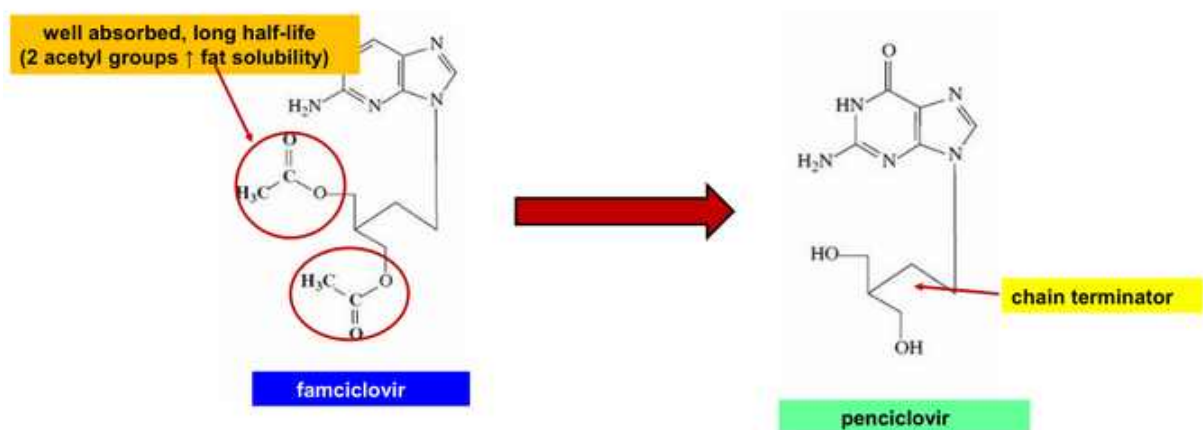
Viral thymidine kinase → acyclovir monophosphate → host kinases → acyclovir triphosphate → viral DNA polymerase incorporates fake guanosine → chain termination.

Valacyclovir



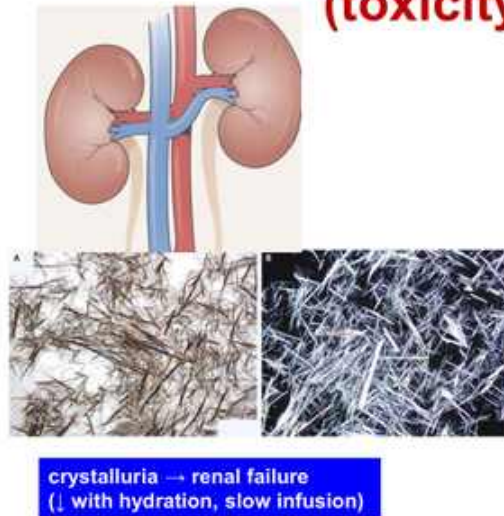
Slide 16. Valacyclovir is the valine ester of acyclovir. The lecture emphasizes improved intestinal absorption and conversion back to acyclovir in blood.

Famciclovir



Slide 17. Famciclovir is converted to penciclovir; the slide emphasizes better absorption/longer half-life and chain termination.

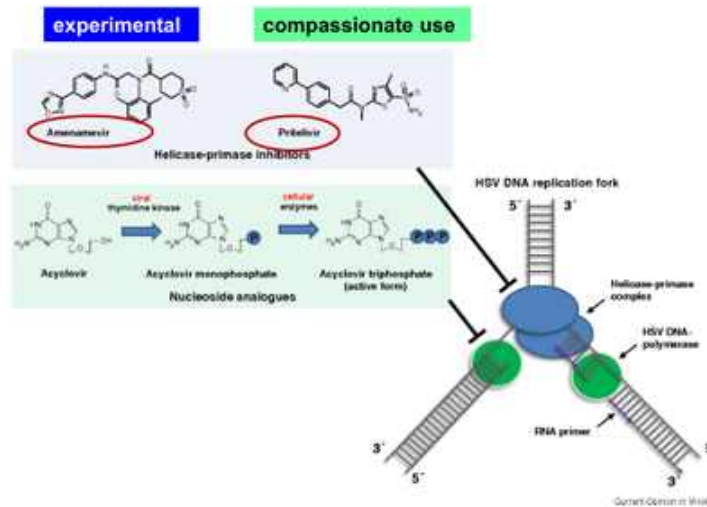
Acyclovir (toxicity)



Slide 19. Acyclovir toxicity: crystalluria leading to renal failure, mitigated by hydration/slow infusion, plus neurotoxicity including agitation, delirium, hallucinations, and myoclonus.

Drug	Current-deck anchor	Major toxicity / pearl
Acyclovir	HSV/VZV guanosine analog; viral TK activation; DNA chain termination	Crystalluria / renal failure; neurotoxicity
Valacyclovir	Valine ester prodrug of acyclovir	Better oral absorption
Famciclovir → penciclovir	Better absorption and long half-life; chain termination	Deck emphasizes structural/ pharmacokinetic relationship

Pritelivir



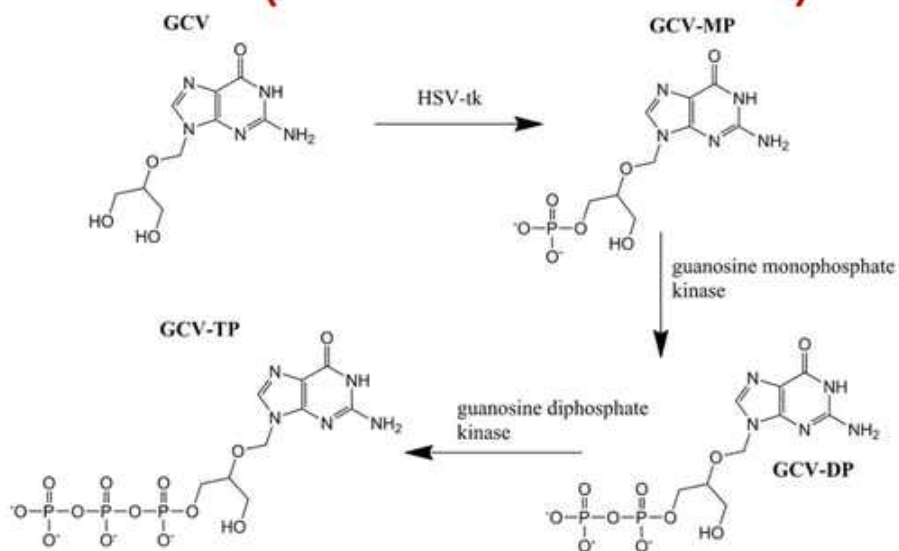
Curr Opin Virol 2016 :18:9-13. doi: 10.1016/j.coviro.2016.01.013.

Slide 20. Pritelivir appears as an experimental/compassionate-use helicase-primase inhibitor, mechanistically distinct from nucleoside analogs.

Deck-only current emphasis: pritelivir is presented as an experimental compassionate-use helicase-primase inhibitor. Do not flatten it into the acyclovir class.

Ganciclovir / Valganciclovir

Ganciclovir (mechanism of action)



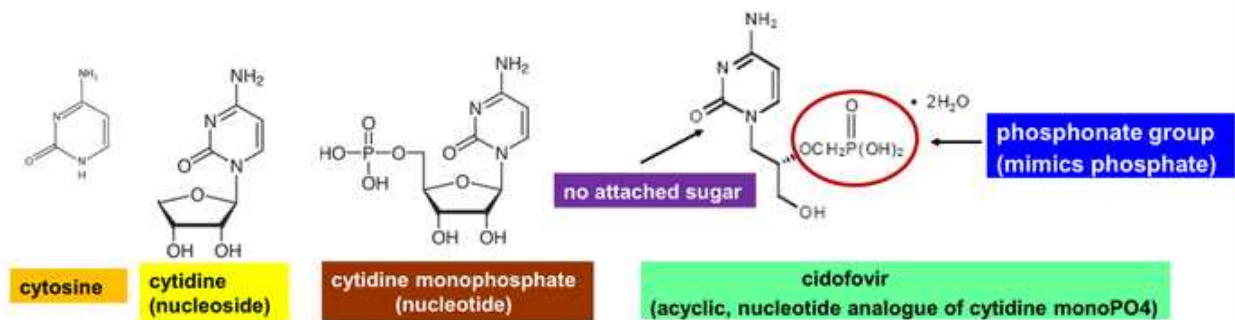
Slide 22. Ganciclovir phosphorylation pathway, followed by host-cell kinase activation.

crystalluria
(less common than acyclovir)

Ganciclovir / valganciclovir = CMV • major toxicity = myelosuppression.

Handout-only activation detail: the companion identifies CMV **UL97 kinase** as the initial activating kinase for ganciclovir and notes UL97 mutation as an important resistance mechanism.

Cidofovir



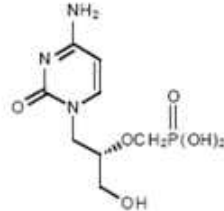
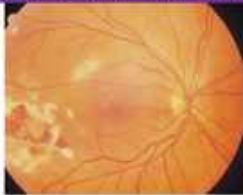
Slide 26. Cidofovir is shown as an acyclic cytidine monophosphate-like nucleotide analog with a phosphonate group that mimics phosphate.

Cidofovir (uses)

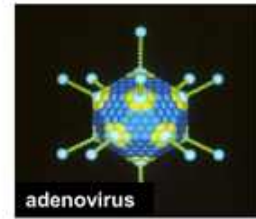
acyclovir-resistant HSV



ganciclovir-resistant CMV infections



cidofovir
(broad-spectrum anti-viral)



adenovirus



human papilloma virus



pox virus

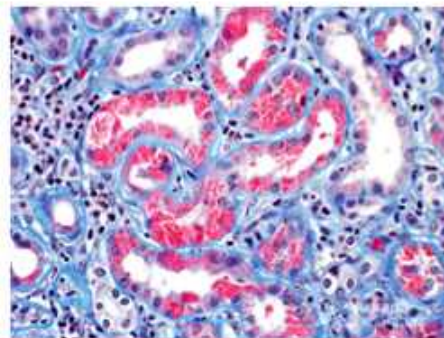
Slide 28. Broad-spectrum uses shown in the current deck: acyclovir-resistant HSV, adenovirus, HPV, poxvirus, and ganciclovir-resistant CMV.

Cidofovir (toxicity)

dose-dependent nephrotoxicity



contraindicated in patients with
Cr > 1.5 or 2+ proteinuria



Fanconi syndrome
(proximal tubule wasting of glucose, HCO₃, aa, phosphates)

Slide 29. Cidofovir toxicity is dose-dependent nephrotoxicity with Fanconi syndrome/proximal tubular wasting; the slide lists renal-function contraindication thresholds.

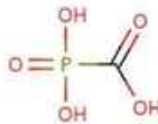
CIDOFOVIR

Nucleotide-like analog → bypasses viral-kinase activation step → resistant herpesvirus option • major problem = proximal-tubule nephrotoxicity / Fanconi syndrome.

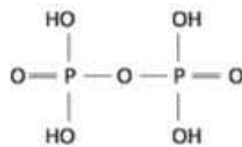
Handout-only management pearl: the handout adds coadministration with **probenecid + aggressive IV hydration** to reduce nephrotoxicity.

Foscarnet

Foscarnet (mechanism of action)



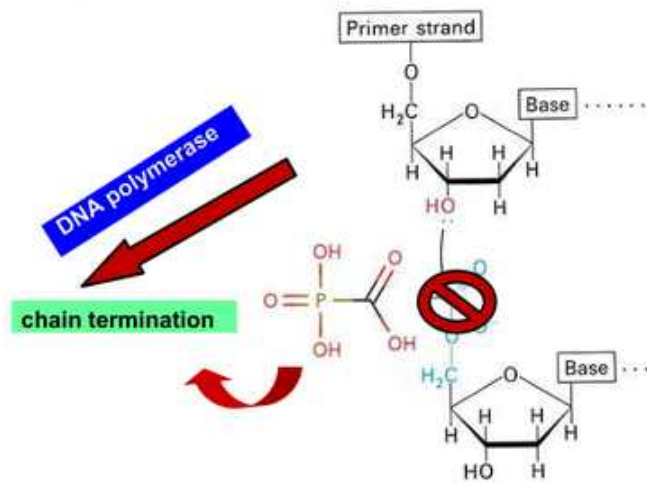
foscarnet
(pyrophosphate analogue)



pyrophosphate

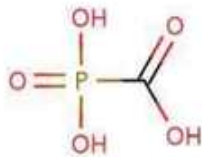
Slide 30. Foscarnet is a pyrophosphate analog.

DNA (formation with foscarnet)



Slide 32. Foscarnet blocks normal pyrophosphate-dependent DNA-polymerase chemistry and produces chain termination without needing a nucleoside-activation pathway.

Foscarnet (uses)



CMV infections when ganciclovir cannot be used



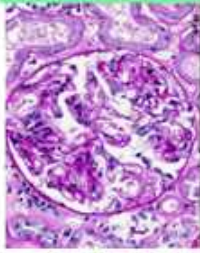
acyclovir-resistant HSV & VZV infections



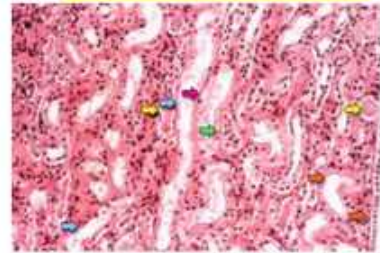
Slide 33. Uses: acyclovir-resistant HSV/VZV and CMV when ganciclovir cannot be used.

Foscarnet (toxicity)

crystal deposition



acute tubular necrosis



nausea with infusion

Slide 34. Current-deck toxicities: crystal deposition, acute tubular necrosis, and nausea with infusion.

Handout-only toxicity enrichment: the handout additionally emphasizes electrolyte disturbances, including hypocalcemia and hypomagnesemia, with seizure risk.

Herpes Resistance Logic

Problem	Why the usual drug fails	Mechanistic escape
Acyclovir-resistant HSV/VZV	Loss/mutation of viral thymidine kinase prevents activation	Foscarnet or cidofovir can bypass that kinase requirement
Ganciclovir resistance / intolerance	Handout emphasizes UL97 kinase mutation; myelosuppression may also limit therapy	Foscarnet or cidofovir depending context

Ozzy checkpoint - click him once you can answer “which herpes drugs need a viral kinase?” without memorizing isolated lists.

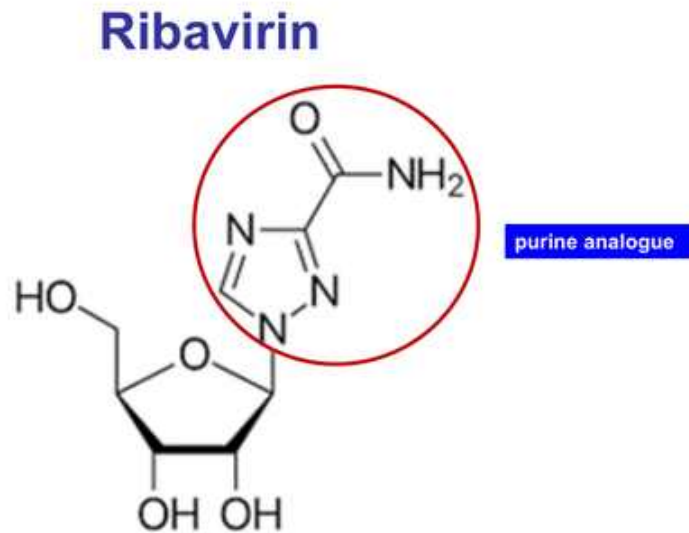
VIRAL KINASE SPLIT

Needs viral kinase: acyclovir; ganciclovir.

Does NOT need viral kinase: foscarnet; cidofovir.

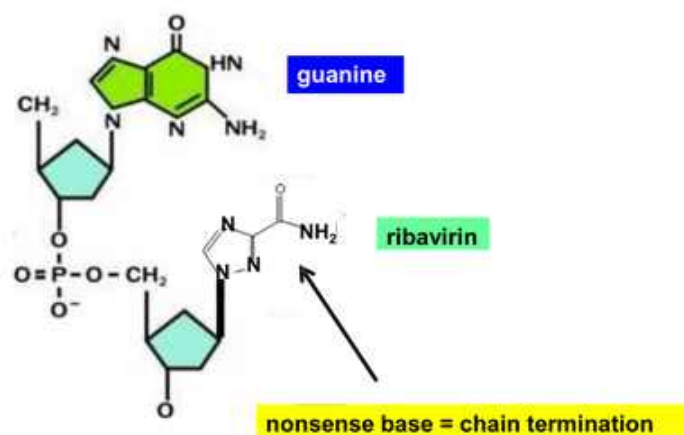
Slides 35-42: Ribavirin & Influenza Agents

Ribavirin



Slide 36. Ribavirin is introduced as a purine analog.

Ribavirin (mechanism of action)

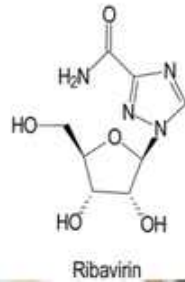


Slide 38. The lecture depicts ribavirin as a 'nonsense base' that disrupts nucleic-acid synthesis and chain elongation.

Ribavirin (uses)



hepatitis C ???



viral hemorrhagic fevers
(eg Lassa fever)



respiratory syncytial virus

Slide 39. Uses shown: RSV, viral hemorrhagic fevers such as Lassa fever, and 'hepatitis C ???' exactly as written on the current slide.

Do not silently clean up slide 39: the current deck literally places “**hepatitis C ???**” under ribavirin uses. Treat that as a questioning/historical cue, not as a confident modern HCV recommendation. The HCV section later teaches modern direct-acting antiviral therapy.

Handout-only toxicity enrichment: ribavirin is associated with **hemolytic anemia** and is a potent **teratogen**.

[illegible]

Slide 41. Neuraminidase-inhibitor toxicity: zanamivir-associated bronchospasm, making it unsuitable for patients with asthma/COPD; confusion is also listed.

(toxicity)



Beloxavir (toxicity)



all rare



Slide 42. Baloxavir toxicities are presented as rare.

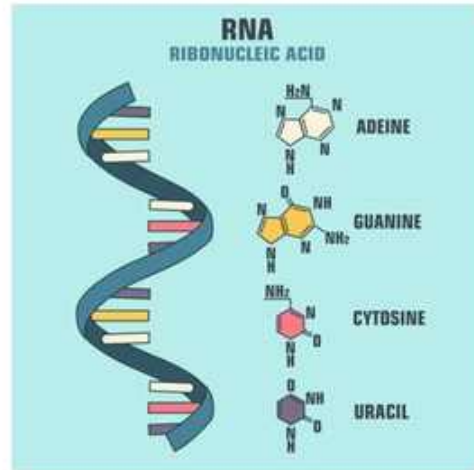
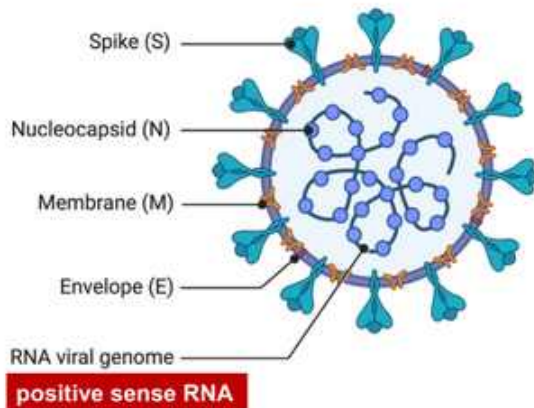
Influenza class	Mechanism	High-yield use / toxicity
Oseltamivir / zanamivir / peramivir	Neuraminidase inhibition → prevents release/emergence of newly formed virions	Influenza A & B; zanamivir can cause bronchospasm
Baloxavir	Handout: cap-dependent endonuclease inhibition / blocks “cap snatching”	Deck: toxicities are rare

Handout-only timing pearl: neuraminidase inhibitors are most effective when started early, classically within 48 hours.

Slides 43-53: COVID-19 Pharmacology

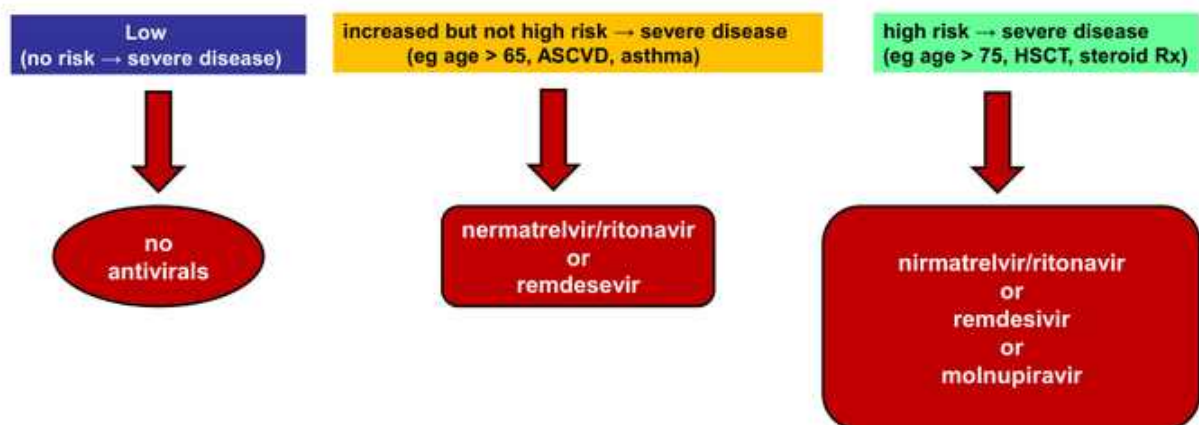
COVID-19

Coronavirus Structure



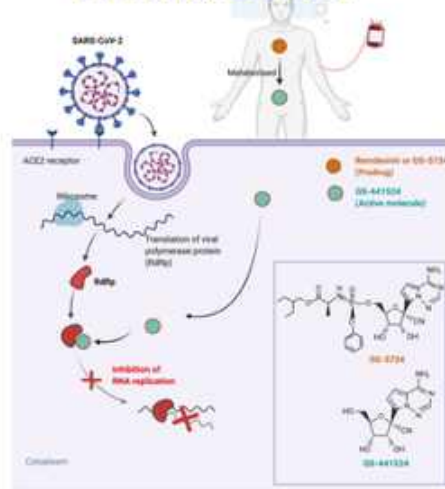
Slide 43. The deck re-establishes SARS-CoV-2 as a positive-sense RNA virus before mapping pharmacology onto its life cycle.

Symptomatic mild-severe COVID-19 (5-7d) (Baseline Patient Risk Assessment)



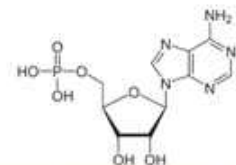
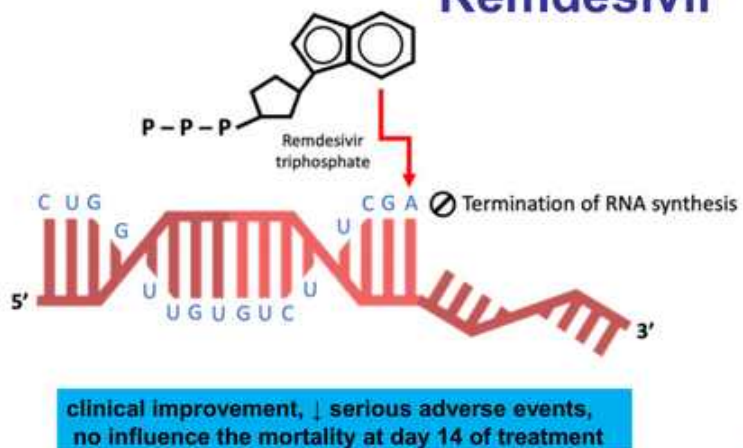
Slide 44. Current 2026 risk-based treatment framework: low-risk mild disease gets no antivirals; increasing risk favors nirmatrelvir/ritonavir or remdesivir, with molnupiravir included for high-risk patients.

Remdesivir

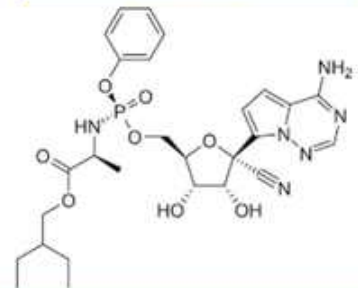


Slide 45. Remdesivir section begins with the drug's role in SARS-CoV-2 treatment.

Remdesivir



adenosine monophosphate (nucleotide)



remdesivir (adenosine monophosphate analogue)

J Pharm Pharm Sci 2021;24:237-45

Slide 46. Remdesivir is shown as an adenosine monophosphate analog.

Remdesivir (role in COVID-19 Rx)

early in illness (≤ 10 days)



require supplemental O₂
(not critically ill)

shortens time to recovery; ↓ risk of progression

Lancet Infect Dis 2021;1:20-21

Slide 47. Current-deck role: early illness, supplemental oxygen but not critically ill; shortens recovery and lowers progression risk.

Remdesivir (toxicity)



infusion-related toxicity
(tachycardia, ↑ or ↓ BP, nausea, rash)



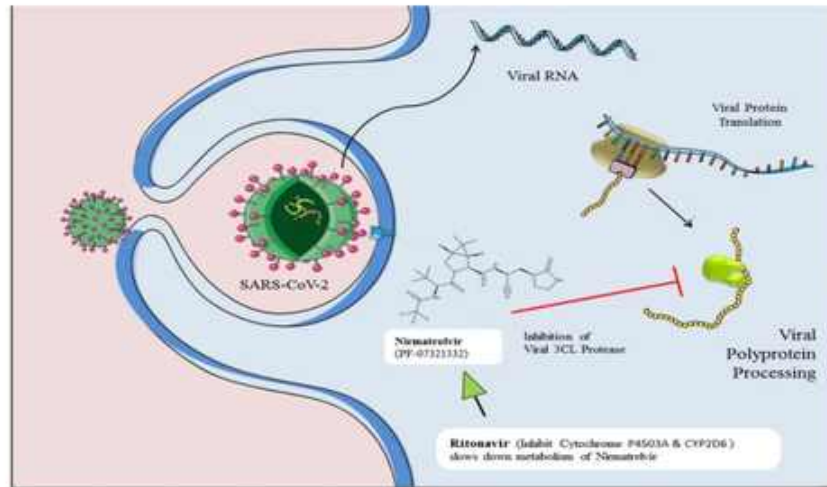
↑ ALT, AST

Slide 48. Remdesivir toxicities: elevated ALT/AST and infusion reactions such as tachycardia, BP changes, nausea, and rash.

REMDESIVIR

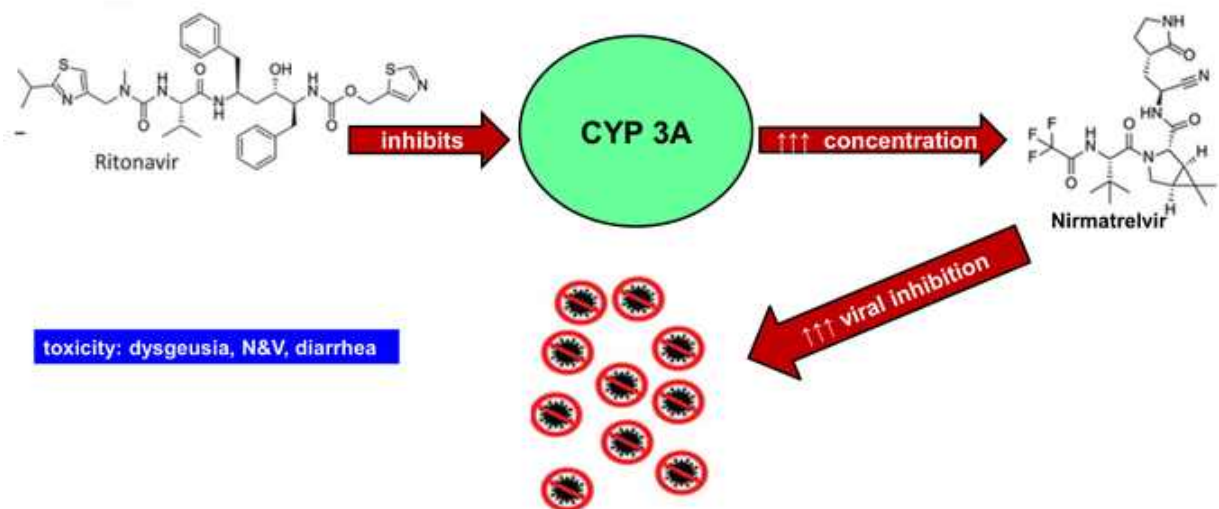
Adenosine nucleotide analog → inhibits viral RNA polymerase / terminates RNA synthesis • watch liver enzymes + infusion reactions.

Nirmatrelvir/ritonavir



Slide 49. Nirmatrelvir/ritonavir treatment pair.

Nirmatrelvir/Ritonavir

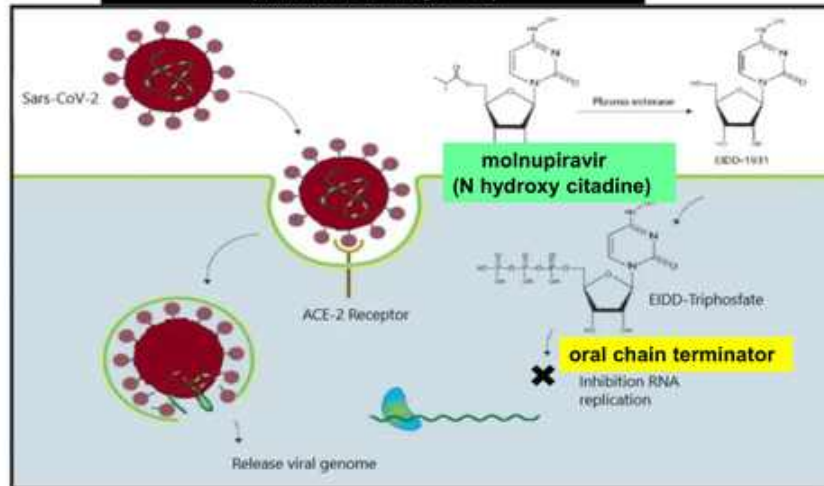


Slide 50. Ritonavir inhibits CYP3A, greatly increasing nirmatrelvir concentration; the deck lists dysgeusia, nausea/vomiting, and diarrhea.

Paxlovid interaction logic: ritonavir is there as a pharmacokinetic booster, so CYP3A drug-drug interactions are a central concern.

Molnupiravir

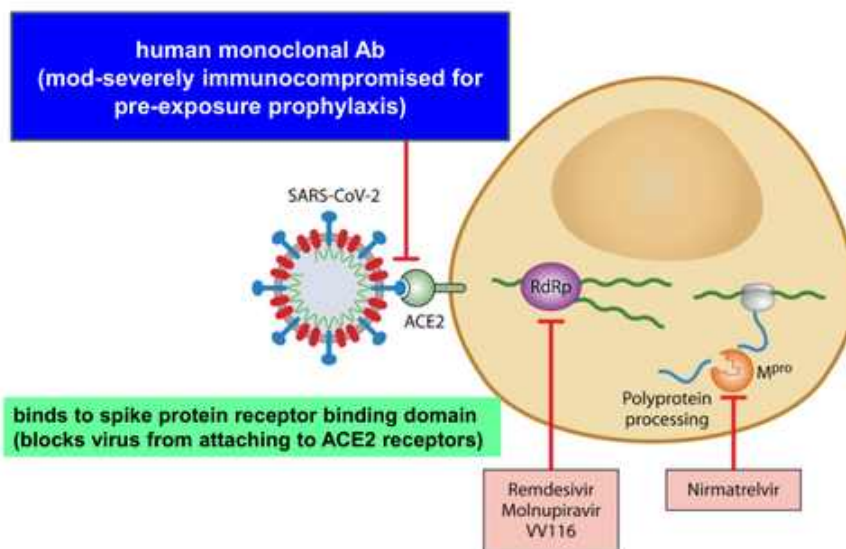
Oral drug for patients at high risk of serious disease
(diabetes, CAD, age > 65)



EIDD = Emory Institute for Drug Development

Slide 51. Molnupiravir is presented as an oral nucleoside-related chain-terminating agent for patients at high risk of serious disease.

Pemivibart

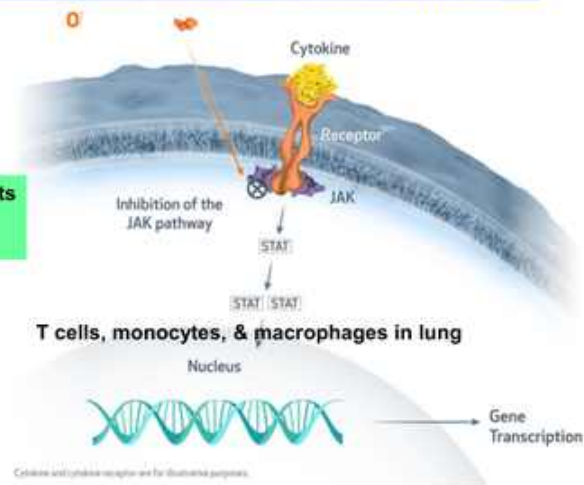


Slide 52. Pemivibart: human monoclonal antibody for pre-exposure prophylaxis in moderately to severely immunocompromised patients; binds the spike RBD and blocks ACE2 attachment.

Baricitinib

blocks Janus kinase enzymes → ↓ inflammatory cytokines

FDA-approved + other Rx for patients requiring supplemental O₂, mechanical ventilation, or ECMO



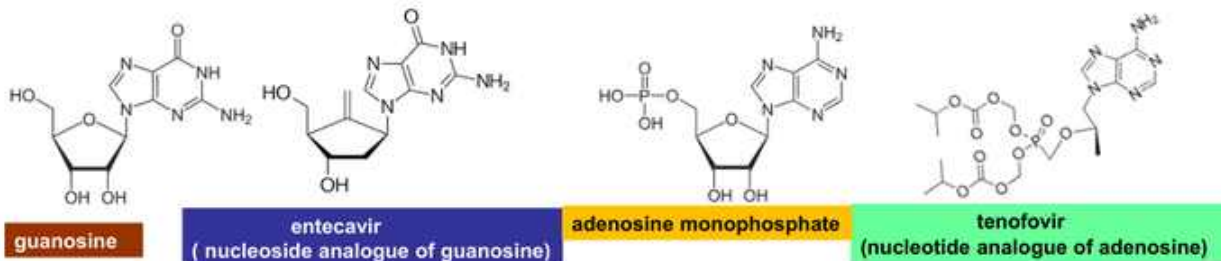
Slide 53. Baricitinib blocks JAK enzymes and reduces inflammatory cytokine signaling; the slide places it in patients requiring supplemental oxygen, mechanical ventilation, or ECMO.

Deck-only distinction: baricitinib is an immunomodulator, not a direct antiviral.

It appears in the antiviral lecture because it is part of COVID-19 treatment strategy, but its mechanism is host JAK inhibition rather than direct inhibition of SARS-CoV-2 replication.

Slides 54-58: Hepatitis B

HBV



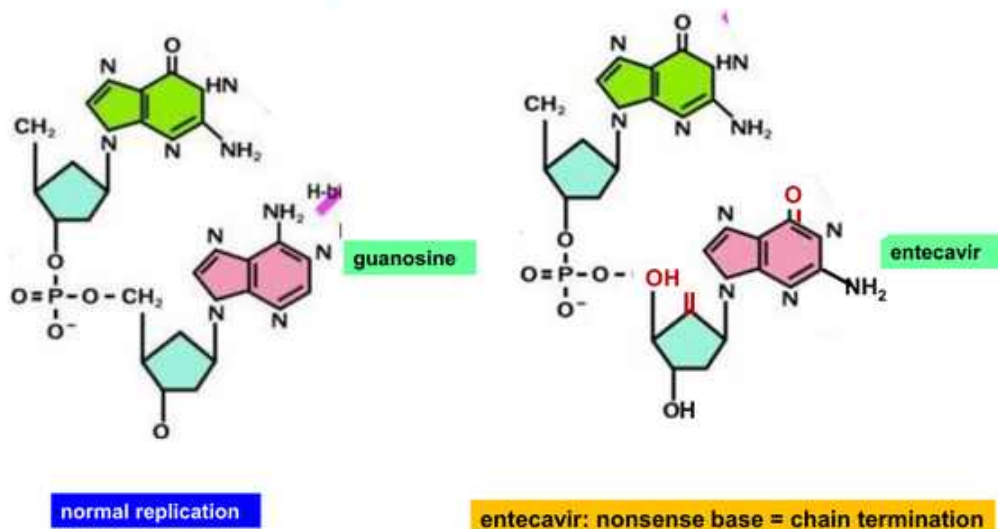
Both highly effective in suppressing HBV viral replication

Lancet 2023;401:1039-52

Slide 54. Current deck: entecavir is a guanosine nucleoside analog and tenofovir an adenosine nucleotide analog; both are described as highly effective at suppressing HBV replication.

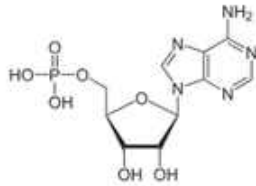
Entecavir (mechanism of action)

side effects:
headache, fatigue, nausea, ↑ liver enz.

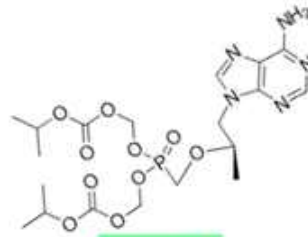


Slide 55. Entecavir acts as a nonsense-base / chain-terminating analog; listed side effects include headache, fatigue, nausea, and increased liver enzymes.

Tenofovir (mechanism of action)



adenosine monophosphate



tenofovir

Tenofovir: nucleotide analogue (NA) of adenosine monophosphate

Lancet 2023;401:1039-52

Slide 56. Tenofovir is shown as an adenosine-monophosphate nucleotide analog.

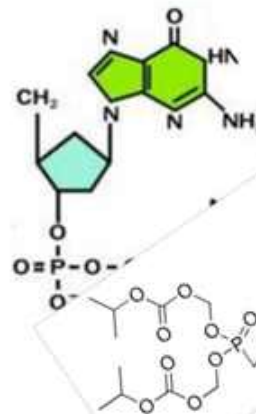
Tenofovir (mechanism of action)

side effects:
headache, N & V, diarrhea
renal impairment (small risk)
↓ bone mineral density



adenosine

normal replication



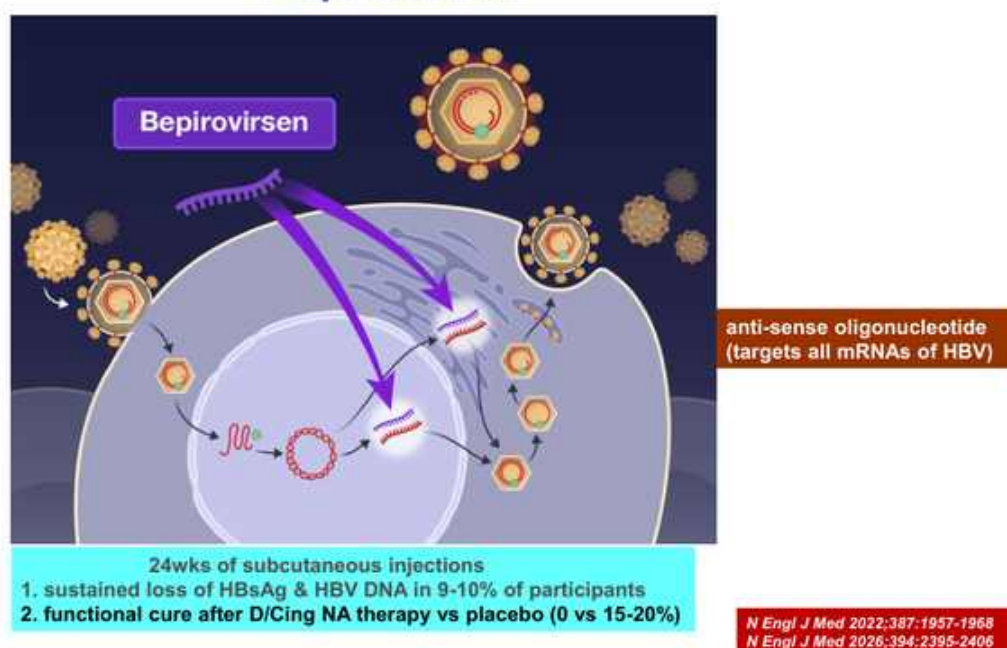
tenofovir

tenofovir: nonsense base = chain termination
long-term use → high rates of viral suppression; no resistance

Slide 57. Tenofovir chain termination; listed adverse effects include GI symptoms, small risk of renal impairment, and decreased bone mineral density.

HBV drug	Mechanism	Current-deck adverse effects
Entecavir	Guanosine nucleoside analog → chain termination	Headache, fatigue, nausea, ↑ liver enzymes
Tenofovir	Adenosine-monophosphate nucleotide analog → chain termination	Headache, nausea/vomiting, diarrhea, small renal risk, ↓ bone mineral density

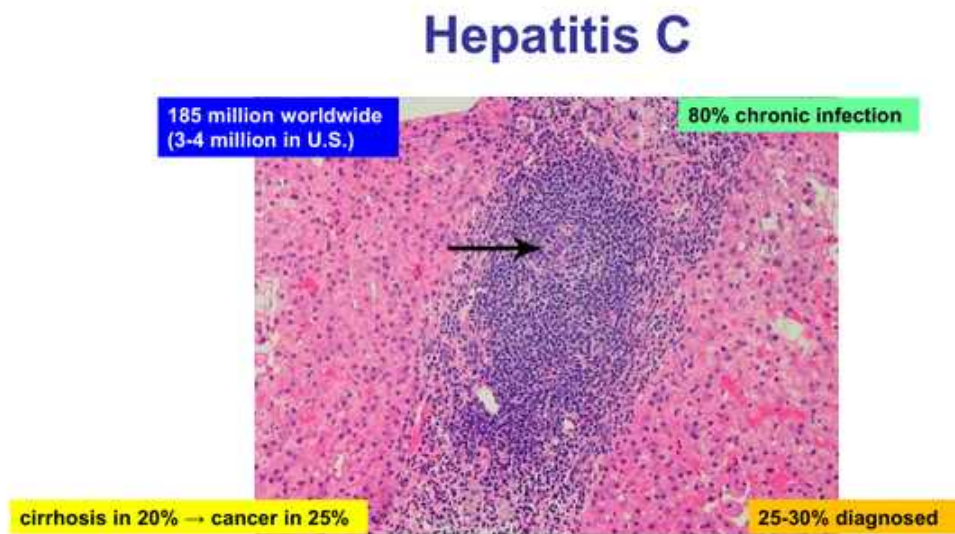
Bepirovirsen



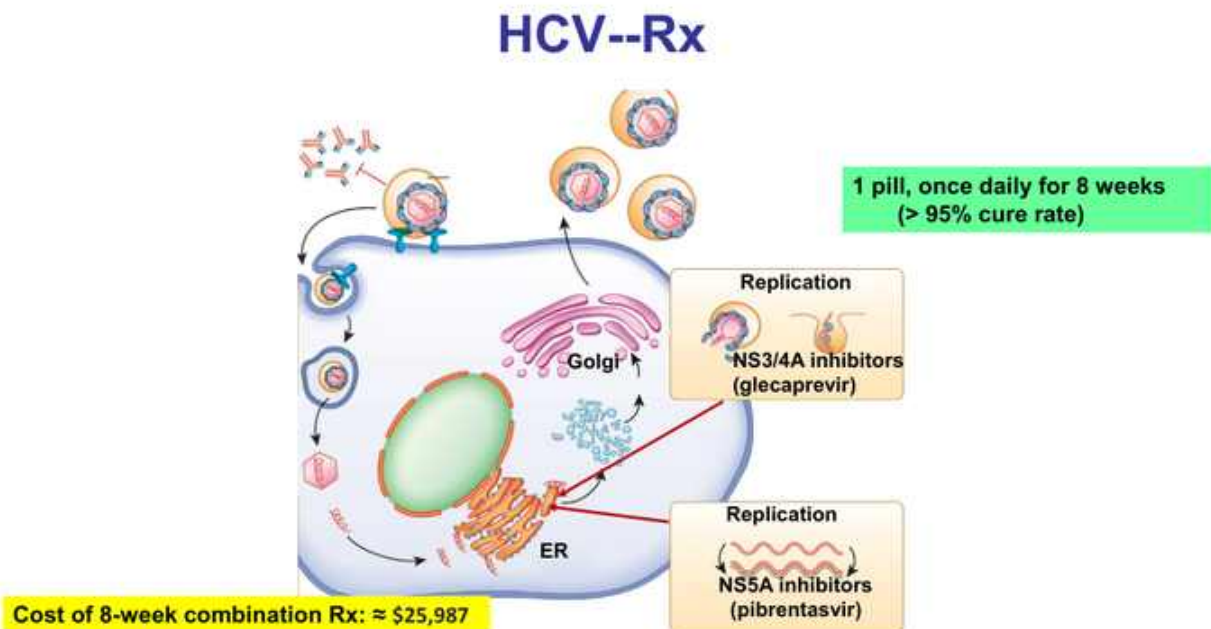
Slide 58. Bepirovirsen is a current-deck 2026 research update: antisense oligonucleotide targeting all HBV mRNAs, with 24 weeks of subcutaneous injections and modest functional-cure outcomes in the cited studies.

Current-deck update: bepirovirsen is presented as an emerging antisense strategy, not established replacement therapy. Keep the study-result context attached to it.

Slides 59-61: Hepatitis C

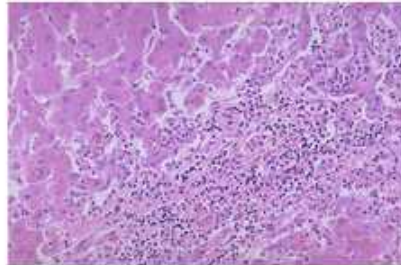


Slide 59. HCV disease burden slide sets up why curative direct-acting therapy matters.



Slide 60. Current-deck 8-week combination: glecaprevir (NS3/4A inhibitor) + pibrentasvir (NS5A inhibitor), one pill daily, with >95% cure rate shown.

HCV drugs (toxicity)



portal inflammation
(esp co-infection with HBV)

Slide 61. Current-deck toxicity warning: portal inflammation, especially in HBV co-infection.

HCV SUFFIX ANCHORS

-previr = protease (NS3/4A) • -asvir = NS5A • handout also adds -buvir = NS5B RNA polymerase.

Handout-only safety framing: modern HCV direct-acting antivirals carry an important warning for **HBV reactivation**; screen for HBV before therapy. The handout also adds NS5B inhibitors such as sofosbuvir, which are not shown in this Fisher slide sequence.

High-Yield Drug-Selection & Toxicity Traps

Vignette / clue	Best answer	Why
HSV/VZV + viral thymidine kinase intact	Acyclovir / valacyclovir	Selective activation in infected cells
Acyclovir-resistant HSV/ VZV	Foscarnet or cidofovir	Bypass viral TK activation
CMV retinitis	Ganciclovir / valganciclovir	Deck emphasizes much greater CMV activity than acyclovir
CMV + severe neutropenia on ganciclovir	Foscarnet	Avoids ganciclovir myelosuppression
Crystalluria / renal failure	Acyclovir	Hydration + slow infusion mitigate risk
Fanconi syndrome	Cidofovir	Proximal-tubule toxicity
Nephrotoxicity + electrolyte abnormalities	Foscarnet	Pyrophosphate analog; handout emphasizes Ca/Mg effects
Influenza A/B prophylaxis/ treatment	Neuraminidase inhibitor	Blocks virion release
Asthma/COPD	Avoid zanamivir	Bronchospasm
High-risk outpatient COVID-19	Nirmatrelvir/ritonavir or remdesivir	Current risk-based slide
HBV suppression	Tenofovir or entecavir	Chain-terminating nucleos(t)ide analogs
HCV 8-week DAA combo	Glecaprevir + pibrentasvir	NS3/4A + NS5A inhibition

LAST 60 SECONDS

Acyclovir = kidney • Ganciclovir = marrow • Cidofovir = Fanconi kidney • Foscarnet = kidney/electrolytes • Zanamivir = bronchospasm • Remdesivir = liver/infusion • Tenofovir = kidney/bone • HCV DAAs = think HBV reactivation screening.